CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Second Semester of M. Tech. Examination (EE)

May 2013

EE 708 Advanced Power Electronics in Power System

Date: 15/05/2013, Wednesday **Time: 10:00 a.m. To 01:00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) Explain briefly working of shunt active power filter. And also compare it with series active power filter. **[04]**

(b) Enlist and discuss the various performance parameters of line commutated converters. **[03]**

Q - 2 (a) A 3-Φ bridge inverter delivers power to a resistive load from a 450 V dc source. For a star connected load of 10 Ω per phase, determine for both (a) 180^{0} mode and (b) 120^{0} mode : **[07]**

- RMS value of load current
- RMS value of thyristor current
- Load power

(b) Explain working of 24 -pulse converter. **[07]**

OR

Q - 2 (a) Explain in detail operation of 3-Φ, 3-level Diode-clamped inverter supplying star connected resistive load. Also explain the role of clamping diodes. **[07]**

(b) Explain in detail advantages of multi-level inverters over conventional two-level inverters. Explain operation of one leg of 3-level Cascaded H-bridge Inverter. **[07]**

Q - 3 (a) Explain with neat current and voltage waveforms rectifier operation of 3-Φ, Full wave diode based current source converter. **[07]**

(b) Explain with a neat block diagram synchronous reference frame theory for generation of reference compensating currents. **[07]**

OR

Q - 3 (a) Explain in detail operation of 3-Φ, Bridge type current source inverter feeding an induction motor. **[07]**

(b) Derive the equation for reference compensating currents using instantaneous reactive power theory and also discuss in brief hysteresis current control technique. **[07]**

SECTION - II

Q - 4 (a) Explain briefly series compensation. **[03]**

(b) Explain briefly working of Unified Power Flow Controller. **[04]**

Q - 5 (a) Explain in detail the objective of shunt compensation when connected at the **[07]** mid-point and at the end of a transmission line.

(b) Explain working of thyristor controlled reactor. Also explain different solutions for reducing the harmonics generated by thyristor controlled reactor. [07]

OR

Q - 5 (a) Explain with neat waveforms working of thyristor switched capacitor. Also explain conditions for transient free switching of thyristor switched capacitor. [07]

(b) Draw operating V-I area for Fixed Capacitor-Thyristor Controlled Reactor (FC-TCR) and Thyristor Switched Capacitor-Thyristor Controlled Reactor (TSC-TCR). Also explain functional control scheme for FC-TCR and TSC-TCR. [07]

Q - 6 (a) Explain in detail operation of GTO Thyristor Controlled Series Capacitor (GCSC). [07]

(b) Explain in detail Overall control structure of Unified Power Flow Controller. [07]

OR

Q - 6 (a) Explain with neat diagram working of static synchronous series compensator. [07]

(b) Draw and explain control scheme for two converter based Interline Power Flow Controller. [07]

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